

PART I – Bandgap Reference

1. Design Consideration

(a) Determine the ratio of the two resistors through calculation, and calculate the expected Vref.

$$I_{D1} = I_{D2} = I_{D3} = \frac{V_T}{R_1} \ln(n)$$

$$V_{REF} = V_{BE3} + \frac{R_2 V_T}{R_1} \ln(n)$$

$$\frac{\partial V_{BE}}{\partial T} = -0.0015 = \frac{\partial \Delta V_{BE}}{\partial T} = \frac{R_2 V_T}{R_1} \ln(n)$$

$$\frac{R_2}{R_1} \ln(n) = 17.2$$

我先取 $R_2/R_1=12.5$ ，則 $n=4$ ，但後來測試之後發現到 $R_2/R_1=18.33$ 左右才會穩定，也就是說我的

$$\frac{R_2}{R_1} \ln(n) = 25.4$$

而

$$\frac{\partial V_{BE}}{\partial T} = \frac{V_{BE} - (4 + m)V_T - E_g/q}{T} = -0.0022$$

由此可知溫度效應的 m 值應該會比講義預測的值還要大，-而 BJT 先假設其 $V_{CE}=0.7$ ，然後先取 2.5μ 來算，可以得出大概是 $V_{REF}=1.17V$ 左右。

$$V_{REF} = V_{CE3} + I_{D3} R_2 = 0.7 + 2.5\mu * 187.5k = 1.17$$

最後的電壓是 $1.36V$ ，誤差為 14.0% ，這是因為由第(b)題可以發現電阻要取比較高，因此電壓最後會上升。

```
+0:v1a    = 1.2836 0:v1b    = 1.2826 0:v1c    = 260.8244m
+0:v1d    = 967.3563m 0:v1e    = 443.8868m 0:v1f    = 447.4432m
+0:v3a    = 1.1474 0:v3b    = 37.7131m 0:v3c    = 1.4547
+0:v3d    = 4.6642m 0:va     = 669.6830m 0:vac     = 0.
+0:vamp_in = 707.7068m 0:vamp_ip = 707.6918m 0:vamp_out = 1.2994
+0:vamp_out = 1.2994 0:vamp_ref = 1.3620 0:vb     = 669.5005m
+0:vbias   = 441.2781m 0:vdd   = 1.8000 0:vref    = 1.3620
+0:vss     = 0.
```

(b) Given the power consumption limitation, arrange the total current you can use carefully; by doing so, you can determine the absolute value of the two resistors, and also be able to design the opamp based on the current budget you set.

我先把電路分成 start-up and constant gm、Amp 和 BGR 三個部分，其中 start-up and constant gm 我能量預計要給 15μ ，Amp 的能量預計要給 15μ 和 BGR 能量預計要給 20μ

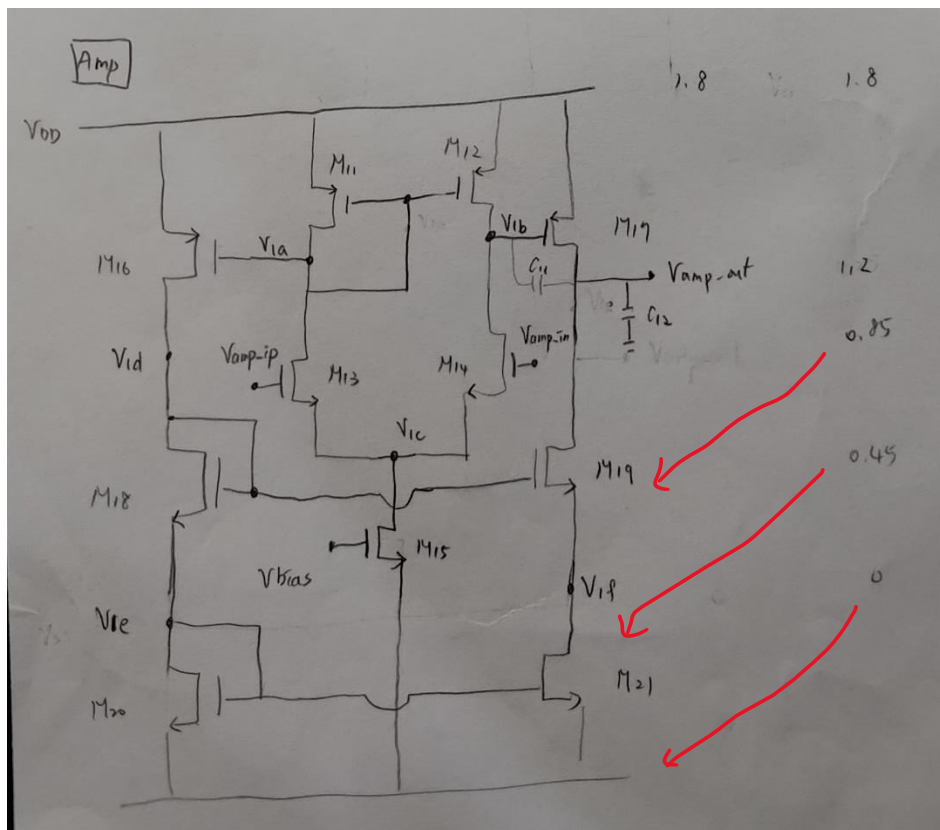
$$I_{D1} = I_{D2} = I_{D3} = \frac{V_T}{R_1} \ln(n)$$

$$3I < \frac{P}{V} = \frac{15\mu}{1.8} = 0.81 * 10^{-5}$$

$$2.7(\mu A) > I = \frac{0.026}{R_1} \ln(4)$$

$$R_1 > 13350(\Omega)$$

R1 先取 15k(Ω)，然後 R2 取 12.5 倍的 187.5k(Ω)，看到圖發現到圖形的現都是由左上往右下傾斜，代表我的正的斜率的比率還不夠高，因此我增加 R2 的電阻直到得到 R2=275k(Ω)時 Bandgap 的部分才比較穩定。



接下來是設計放大器，我選二階放大器是因為如果要有虛擬的短路和 PSR 要夠小，放大器的放大倍率就要高，而外層的放大器因為要讓 output 夠大這樣 BGR 上面的 PMOS 才會進 sat. region，因此我設計下面有兩個 NMOS 上面有一個 PMOS 來提升電壓，這時候我先假設 Vbias=0.45V，start-up and constant gm 之後再設計，一開始我會先假設我要設計的電壓如上圖的右邊，而 L 因為要降低 channel length modulation 因此先把它固定在 1u，M15 電流源的部分我先把牠設計程 2u 左右，以免 power 太大，再把 V1a 和 V1b 設計到 1.2V 左右，因為 V1a 在 M18 中可以看到 Gate 和 Drain 是接在一起而因為電路是對稱的，因此 Vout

電壓不會太低，這樣 BGR 的 M1M2M3 才會比較容易在 sat. region，接下來放大器的 W 故意不要設太大，這樣放大倍率才會高，而 PMOS 比 NMOS 的 W 大概為 3 到 5 倍是因為 PMOS 本來就比 NMOS 不容易留電流了，而且我 Vout 的電壓想要大一點，因此才會有這樣的倍率。下面是最後測量的數據。

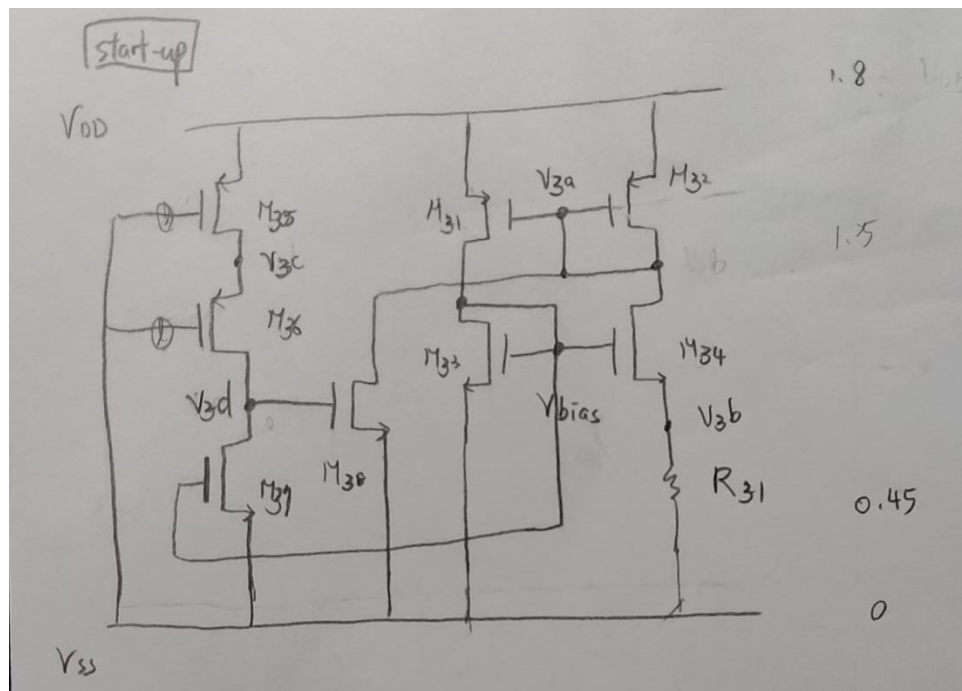
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****      small-signal transfer characteristics
v(vamp_out)/vdiff      = 24.4790k
input resistance at vdiff      = 1.000e+20
output resistance at v(vamp_out) = 31.0723x
```

(c) PSR (Power Supply Rejection) is an important indicator for evaluating the quality of a bandgap reference circuit. Please analyze the aspects through which PSR can be improved and how the gain of the opamp would affect PSR. Use proper calculation and analysis to establish the specifications for the opamp.

$$PSR = \frac{R_2}{A_V R_1} < -40(db) = 0.01$$

由上述公式可以觀察到 AV 要大於 1833 倍，但是我最後設計 24479 倍，這樣 PSR 就可以壓更小，而且我們要考慮不同溫度對 PSR 所造成的影響，因此才會想取那麼大，另外再 PSR 還可以受到 BGR 上面 M1M2M3 三個 PMOS 中的 W 來改變，如果我把 W 設的很大 MOS 中的 Threshold voltage mismatch 會下降，這樣可以降低 Vdd 對電流的影響，因此最後會對 Vref 的影響變化下降。

(d) Design the opamp based on the structure shown in Fig. 3. and Fig. 4. and also the analysis you did in the previous parts.



我先設計 Amp 的電路，然後再設計 start-up and constant gm 的電路，而 start-up and constant gm 的部分，我先設計 constant gm 的部分，我由公式

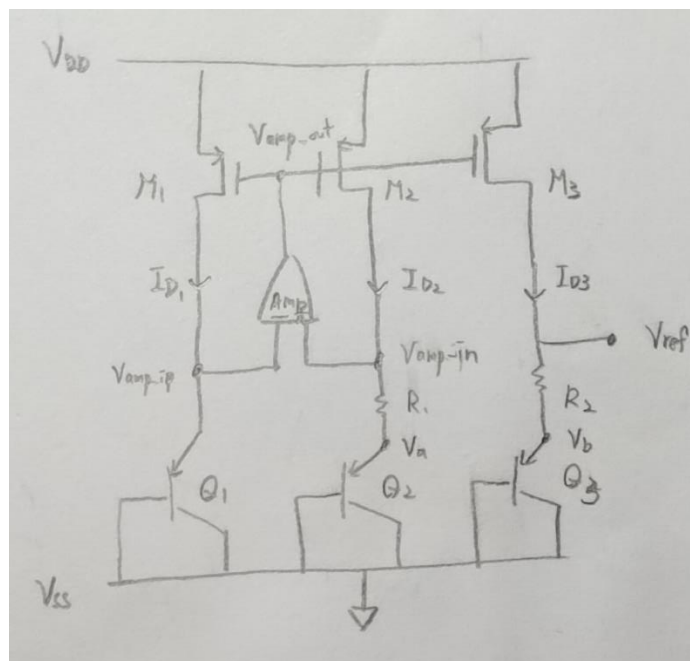
$$I_{D1} = \frac{2}{\mu_n C_{ox} K \left(\frac{W}{L}\right)_N} \frac{1}{R_S^2} \left(1 - \frac{1}{\sqrt{K}}\right)^2$$

來發現到如果我要降 power，我電流要下降，這樣我 K 不能太大，因此我選 K=2，因為 K 不能是 1，而(W/L)大一點也可以降電流，因此我選(W/L)=(5u/1u)，最後 RS 再由電流調整 RS 值，最後是取 8K，至於 start-up 的部分，我是先分析電路發現到其中在 start-up 穩定後 M35 會是 linear. Region、M36 會是 sat. region、M37 會是 linear. region 和 M38 會是 cutoff. Region，這是因為在穩定後 M38 的 MOS 電流應該要斷掉，因此其電流越小越好，所以我的 M38 的 L 特別小來提升 Vth，而要關掉 M38 其中的 V3d 點電壓就不能太大，這樣 M38 的 Vgs 才不會太大而打開，因此 M37 的 Vds 的電壓要很小，所以我的 W 特別大，增加 M37 的電流，來讓 V3d 下降，因為這會導致 M36 的 Vds 會很大而進去 sat. region，而 M35 因為 Vgs 的電壓太大，因此會進去 linear. Region，最後在控制 M37 讓 Vbias 的電壓盡量維持在當初 Amp.設計的 0.45V，而最後我設計為 0.44V，這會導致 M15 的 Vgs 下降，而導致其電流下降，最後測量單純 Amp 的

power=2.2u 而 start-up and constant gm 的部份的 power=9.8u，都在預期規劃的 power 以下。

subckt	0:m35		0:m36		0:m37		0:m38	
element	0:p_18.1		0:p_18.1		0:n_18.1		0:n_18.1	
model	Linear		Saturation		Linear		Cutoff	
region	id		ibs		ibd		vgs	
	-645.0727n	-645.0722n	645.0755n	696.5454f				
	4.093e-22	26.3357a	-1.350e-22	-7.646e-28				
	26.3349a	136.9228a	-18.7734a	-228.7516a				
	-1.8000	-1.4547	441.2781m	4.6642m				
	-345.3082m	-1.4500	4.6642m	1.1474				
	0.	345.3082m	0.	0.				
	-452.3510m	-542.2082m	315.4648m	448.0537m				
	-1.3530	-902.0463m	126.1564m	43.1078m				
	-1.3476	-912.4837m	125.8132m	-443.3895m				
	1.6092u	1.5656u	1.1820m	636.4289u				
	557.0846m	554.7649m	507.4459m	507.4459m				
	528.9176n	1.3881u	5.0327u	25.2071p				
	1.5925u	3.0788n	135.5422u	305.6423f				
	242.9986n	411.2686n	1.0613u	5.5525p				
	16.1560f	399.5146a	2.9944p	1.3405f				
	20.2215f	14.9441f	3.2784p	1.6859f				
	19.0940f	17.2089f	3.0190p	1.6997f				
	6.5335f	5.2973f	886.3441f	3.1728f				
	11.3676f	13.4338f	1.9090p	388.3281a				

(e) Complete the whole bandgap reference circuit design.



接下來我把設計 **start-up and constant gm** 和 **Amp** 接在一起，發現到 **MOS** 都在我預期之中，而接著我把 **Amp** 接到 **BGR**，讓整個迴路成為一個穩定的 **feedback**，而因為最後要降低 **PAR**，因此我在設計 **M1**、**M2** 和 **M3** 時把 **W/L** 中的 **W** 設計大一點，這樣 **PAR** 才會比較小，而 **L** 的部分因為要當電壓對 **Bandgap Reference** 中的溫度不要互相影響太大，也就是圖中的電壓峰值不要離的太遠，因此我把 **L** 設計得比較大來降低其 **channel length modulation**。

下面是一些元件在 TT 27°C 的資訊

element	0:q1	0:q2	0:q3
model	0:pnv_v50x	0:pnv_v50x	0:pnv_v50x
region	FwdActive	FwdActive	FwdActive
ib	-1.0691u	-1.0683u	-1.0612u
ic	-1.4658u	-1.4666u	-1.4568u
vbe	-707.6918m	-669.6830m	-669.5005m
vce	-707.6918m	-669.6830m	-669.5005m
vbc	0.	0.	0.
vs	-268.7669u	-70.6574u	-70.1962u
power	1.7939u	1.6976u	1.6858u
betad	1.3711	1.3728	1.3728
gm	53.7035u	53.7602u	53.4027u
rpi	25.4987k	25.5171k	25.6881k
rx	251.3972	66.1385	66.1470
ro	530.6661x	530.7574x	534.3128x
cpi	39.2158f	154.5722f	154.5611f
cmu	28.0877f	112.3870f	112.3871f
cbx	0.	0.	0.
ccs	0.	0.	0.
betaac	1.3694	1.3718	1.3718
ft	126.9947x	32.0505x	31.8387x

subckt						
element	0:m1	0:m2	0:m3	0:m11	0:m12	0:m13
model	0:p_18.1	0:p_18.1	0:p_18.1	0:p_18.1	0:p_18.1	0:n_18.1
region	Saturation	Saturation	Saturation	Saturation	Saturation	Saturation
id	-2.5349u	-2.5349u	-2.5181u	-602.8989n	-602.9374n	602.8976n
ibs	3.243e-22	3.243e-22	3.222e-22	9.335e-23	9.335e-23	-64.7905a
ibd	6.3832f	6.3831f	2.5598f	172.9730a	173.3179a	-318.8558a
vgs	-500.6267m	-500.6267m	-500.6267m	-516.3910m	-516.3910m	446.8674m
vds	-1.0923	-1.0923	-438.0344m	-516.3910m	-517.4207m	1.0228
vbs	0.	0.	0.	0.	0.	-260.8244m
vth	-464.4684m	-464.4684m	-464.4684m	-494.4815m	-494.4815m	441.3183m
vdsat	-77.8103m	-77.8103m	-77.8103m	-77.6175m	-77.6175m	67.6410m
vod	-36.1583m	-36.1583m	-36.1583m	-21.9095m	-21.9095m	5.5491m
beta	1.1685m	1.1685m	1.1685m	321.3359u	321.3359u	460.8589u
gam_eff	557.0847m	557.0847m	557.0847m	557.0847m	557.0847m	514.2865m
gm	45.8294u	45.8294u	45.5437u	11.2535u	11.2542u	12.7199u
gds	20.8001n	20.8002n	38.3266n	37.4223n	37.3862n	147.3770n
gmb	14.0269u	14.0269u	13.9381u	3.3717u	3.3719u	2.2420u
cdtot	101.6494f	101.6496f	115.6041f	5.4415f	5.4403f	1.9243f
cgtot	3.0154p	3.0154p	3.0184p	22.9944f	22.9944f	7.1149f
cstot	3.1015p	3.1015p	3.0992p	24.8165f	24.8165f	7.2857f
cbtot	1.3628p	1.3628p	1.3754p	16.8638f	16.8626f	5.1892f
cgs	2.4854p	2.4854p	2.4885p	17.3180f	17.3179f	5.1979f
cgd	34.1343f	34.1343f	34.9665f	1.6172f	1.6172f	549.0614a

subckt						
element	0:m14	0:m15	0:m16	0:m17	0:m18	0:m19
model	0:n_18.1	0:n_18.1	0:p_18.1	0:p_18.1	0:n_18.1	0:n_18.1
region	Saturation	Saturation	Saturation	Saturation	Saturation	Saturation
id	602.9361n	1.2058u	-613.5091n	-613.9724n	613.5093n	613.9713n
ibs	-64.7905a	-4.154e-22	9.499e-23	9.506e-23	-79.7846a	-80.4238a
ibd	-318.6000a	-64.7899a	278.9067a	167.6925a	-173.8725a	-233.5494a
vgs	446.8824m	441.2781m	-516.3910m	-517.4207m	523.4694m	519.9130m
vds	1.0218	260.8244m	-832.6437m	-500.6267m	523.4694m	851.9301m
vbs	-260.8244m	0.	0.	0.	-443.8868m	-447.4432m
vth	441.3272m	396.3533m	-494.4815m	-494.4815m	484.5711m	482.0468m
vdsat	67.6435m	84.3851m	-77.6191m	-78.1061m	85.2659m	84.7218m
vod	5.5552m	44.9248m	-21.9095m	-22.9392m	38.8983m	37.8663m
beta	460.8591u	458.8765u	321.3357u	321.2788u	244.7718u	244.7779u
gam_eff	514.2865m	507.4460m	557.0847m	557.0847m	518.6300m	518.7113m
gm	12.7206u	21.8313u	11.4384u	11.4201u	11.4713u	11.4932u
gds	147.3912n	314.1019n	30.8007n	38.7082n	149.4626n	143.2753n
gmb	2.2422u	4.5155u	3.4283u	3.4214u	1.8454u	1.8301u
cdtot	1.9245f	2.3213f	5.1302f	5.4603f	1.1229f	1.0728f
cgtot	7.1153f	9.2697f	22.9863f	23.1183f	4.7376f	4.7130f
cstot	7.2862f	10.2472f	24.8248f	25.0046f	5.1035f	5.0710f
cbtot	5.1894f	6.0350f	16.5559f	16.8869f	2.7977f	2.7468f
cgs	5.1984f	7.6821f	17.3096f	17.4799f	3.9355f	3.9023f
cgd	549.0581a	548.3941a	1.6151f	1.6175f	282.2027a	281.9101a

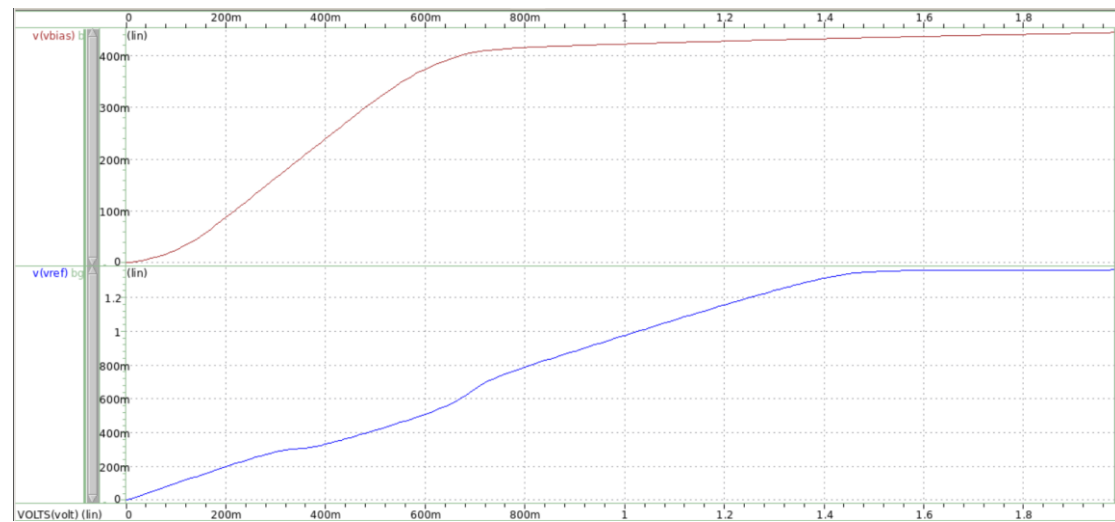
subckt	0:m20	0:m21	0:m31	0:m32	0:m33	0:m34
element	0:n_18.1	0:n_18.1	0:p_18.1	0:p_18.1	0:n_18.1	0:n_18.1
model	Saturation	Saturation	Saturation	Saturation	Saturation	Saturation
region	Saturation	Saturation	Saturation	Saturation	Saturation	Saturation
id	613.5085n	613.9708n	-4.8457u	-4.7141u	4.8457u	4.7141u
ibs	-2.867e-22	-2.869e-22	7.366e-22	7.166e-22	-1.193e-21	-44.6334a
ibd	-79.7840a	-80.4233a	496.4770a	238.4467a	-261.1180a	-1.3580f
vgs	443.8868m	443.8868m	-652.5645m	-652.5645m	441.2781m	403.5649m
vds	443.8868m	447.4432m	-1.3587	-652.5645m	441.2781m	1.1097
vbs	0.	0.	0.	0.	0.	-37.7131m
vth	402.4819m	402.4548m	-494.1037m	-494.1036m	386.9450m	389.7510m
vdsat	83.0664m	83.0807m	-166.0104m	-166.0032m	89.5334m	69.5675m
vod	41.4050m	41.4320m	-158.4609m	-158.4609m	54.3331m	13.8139m
beta	243.0672u	243.0669u	346.8428u	346.8436u	1.5365m	3.0711m
gam_eff	507.4460m	507.4460m	557.0846m	557.0846m	507.4460m	508.4922m
gm	11.2619u	11.2688u	49.6348u	48.3865u	84.1764u	96.1370u
gds	130.0608n	129.9177n	153.4538n	241.8499n	985.1579n	1.0206u
gmb	2.3178u	2.3190u	15.0627u	14.6766u	17.3281u	19.2552u
cdtot	1.2334f	1.2324f	5.2901f	5.8856f	7.0773f	12.5578f
cgtot	4.8616f	4.8621f	32.3966f	32.4149f	31.7877f	51.6859f
cstot	5.4421f	5.4429f	38.0731f	38.0597f	35.1372f	54.1271f
cbtot	3.2891f	3.2882f	18.1311f	18.7096f	19.1930f	36.2435f
cgs	3.9954f	3.9959f	28.2703f	28.2851f	26.7163f	38.8668f
cgd	288.8105a	288.7837a	1.7944f	1.8029f	1.7970f	3.6620f

subckt	0:m35	0:m36	0:m37	0:m38
element	0:p_18.1	0:p_18.1	0:n_18.1	0:n_18.1
model	Linear	Saturation	Linear	Cutoff
region	Linear	Saturation	Linear	Cutoff
id	-645.0727n	-645.0722n	645.0755n	696.5454f
ibs	4.093e-22	26.3357a	-1.350e-22	-7.646e-28
ibd	26.3349a	136.9228a	-18.7734a	-228.7516a
vgs	-1.8000	-1.4547	441.2781m	4.6642m
vds	-345.3082m	-1.4500	4.6642m	1.1474
vbs	0.	345.3082m	0.	0.
vth	-452.3510m	-542.2082m	315.4648m	448.0537m
vdsat	-1.3530	-902.0463m	126.1564m	43.1078m
vod	-1.3476	-912.4837m	125.8132m	-443.3895m
beta	1.6092u	1.5656u	1.1820m	636.4289u
gam_eff	557.0846m	554.7649m	507.4459m	507.4459m
gm	528.9176n	1.3881u	5.0327u	25.2071p
gds	1.5925u	3.0788n	135.5422u	305.6423f
gmb	242.9986n	411.2686n	1.0613u	5.5525p
cdtot	16.1560f	399.5146a	2.9944p	1.3405f
cgtot	20.2215f	14.9441f	3.2784p	1.6859f
cstot	19.0940f	17.2089f	3.0190p	1.6997f
cbtot	6.5335f	5.2973f	886.3441f	3.1728f
cgs	11.3676f	13.4338f	1.9090p	388.3281a

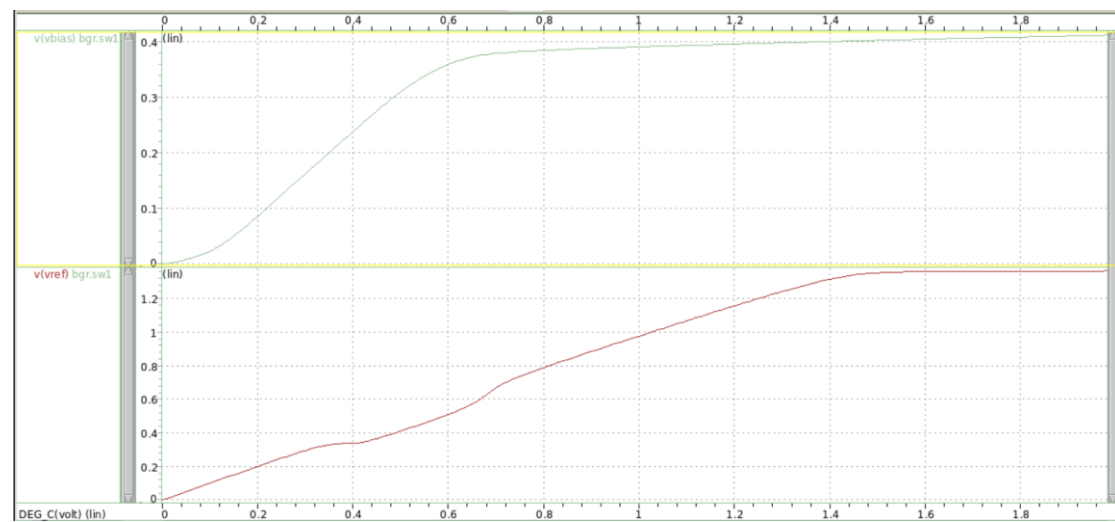
2. Simulation Result

(a) .DC

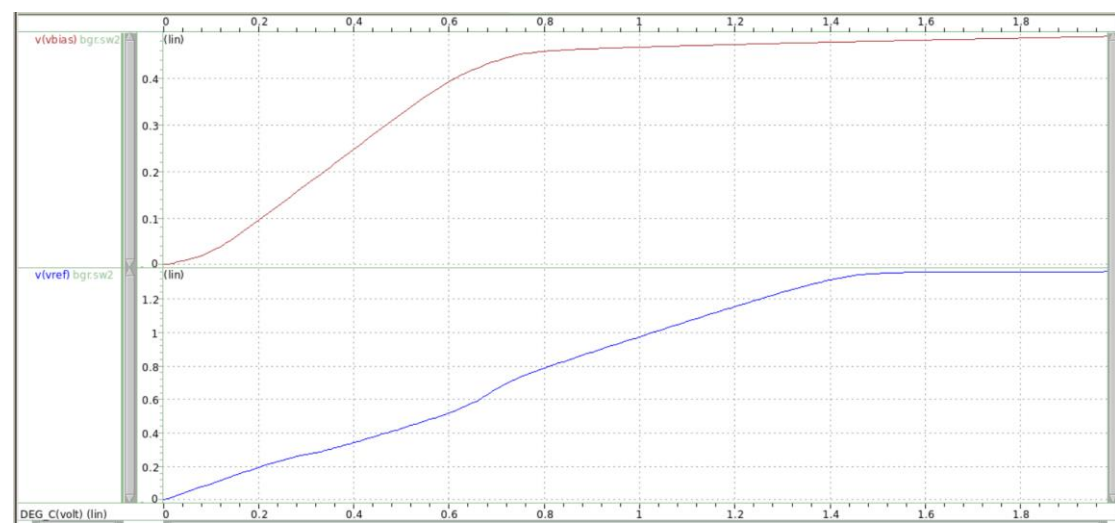
TT



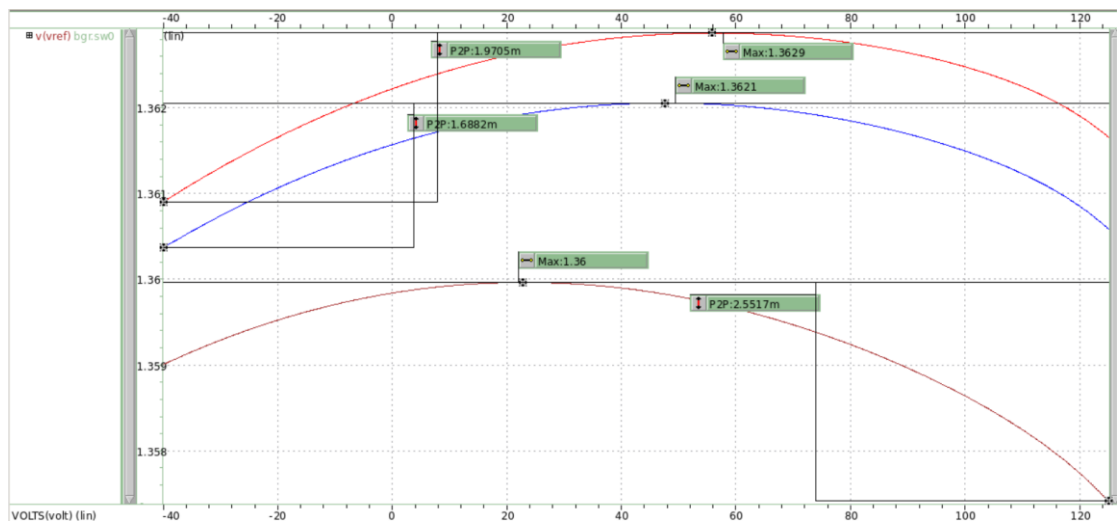
FF



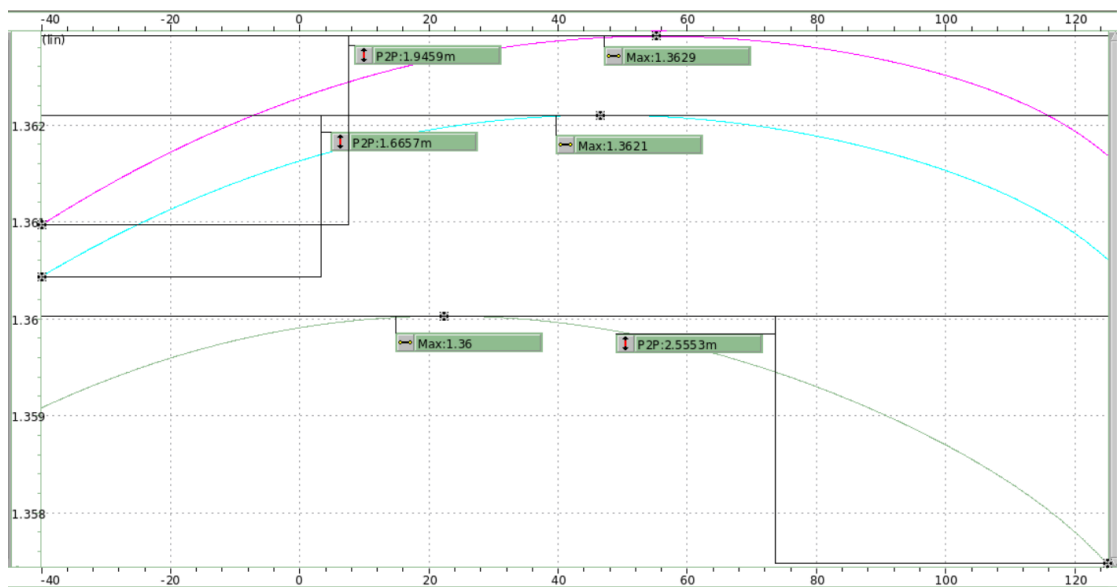
SS



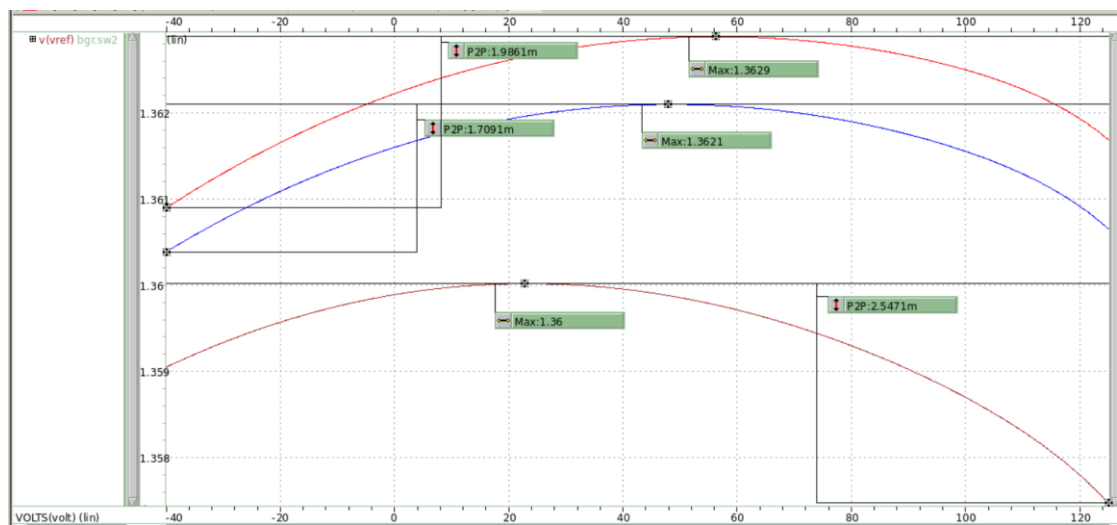
TT



FF

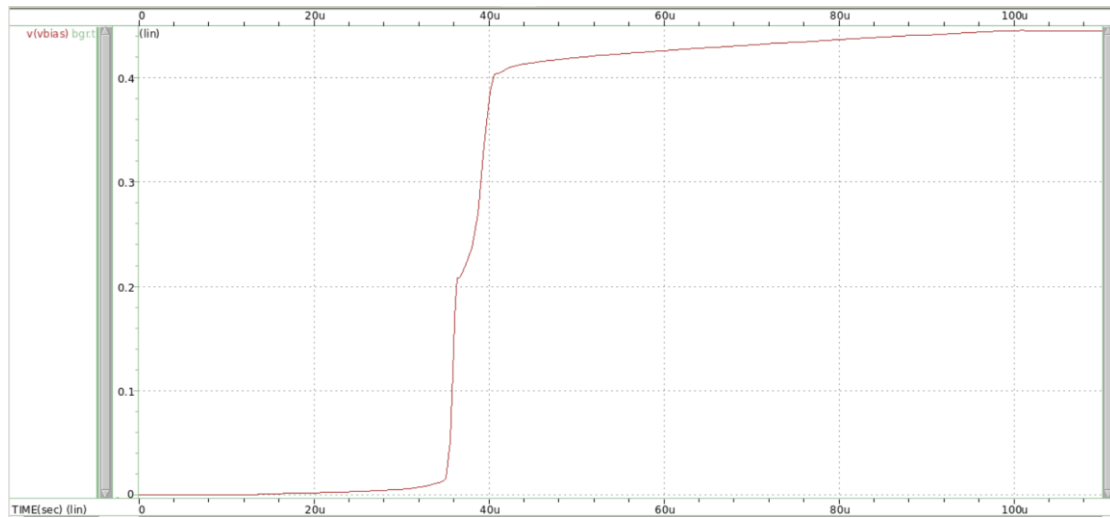


SS

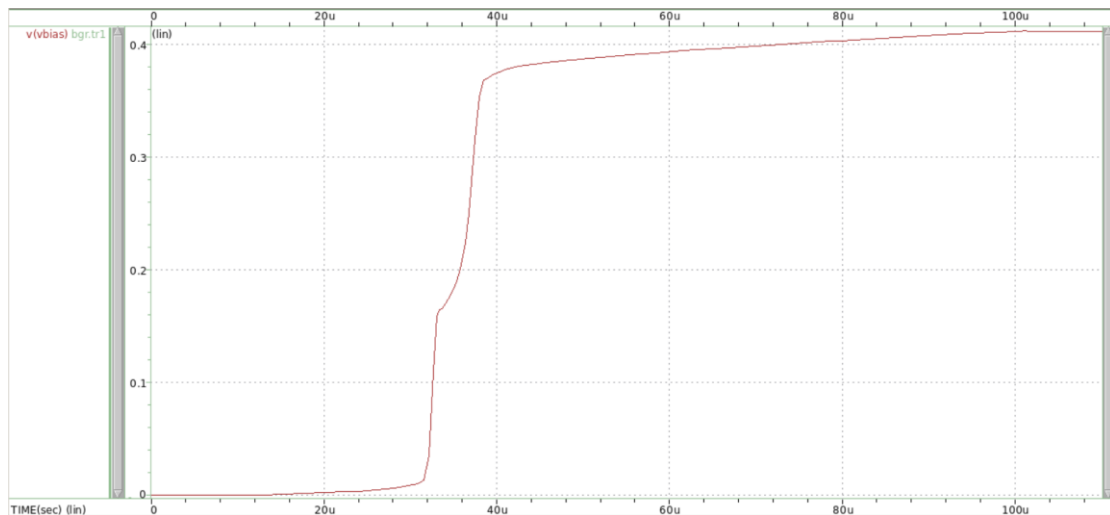


(b) .Tran

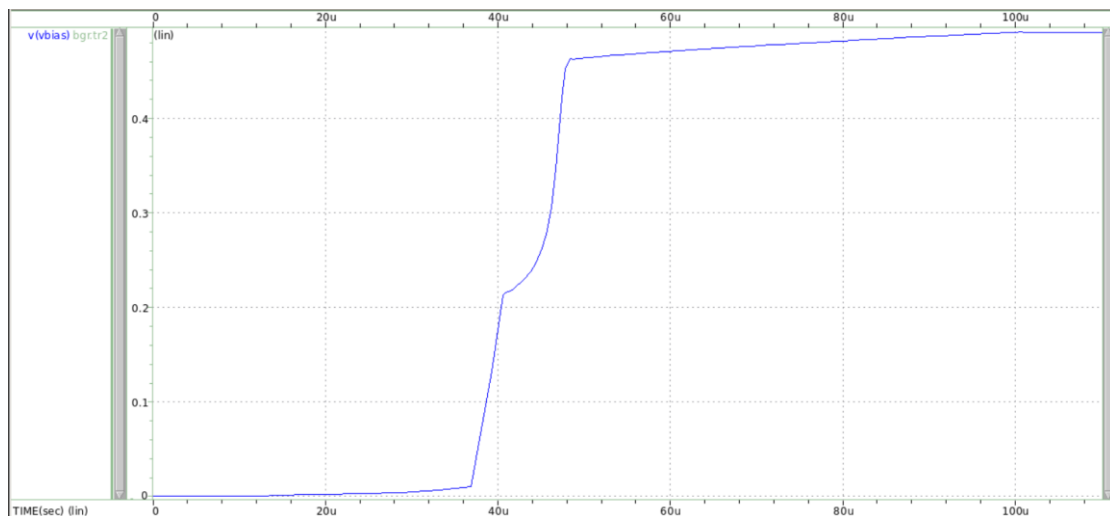
TT



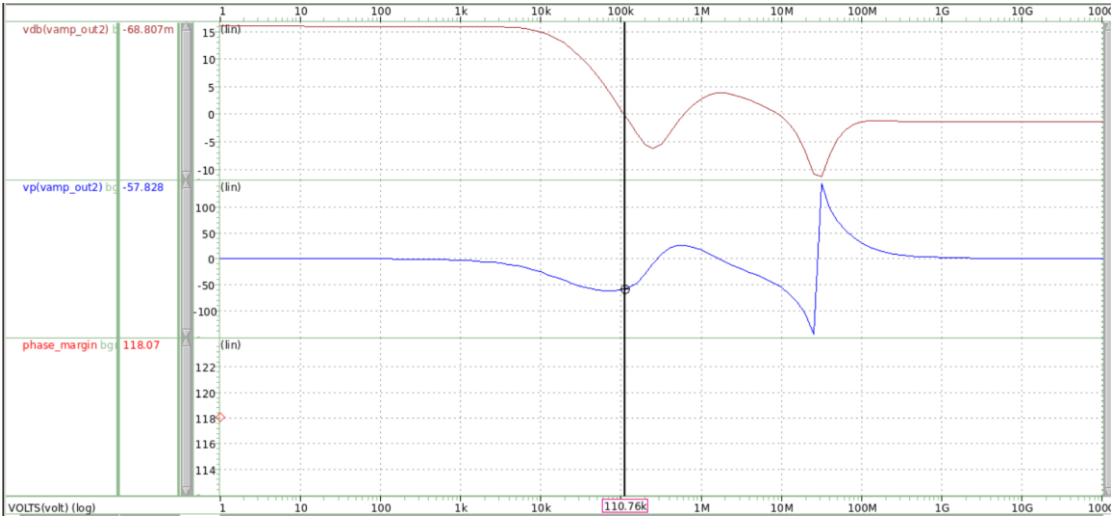
FF



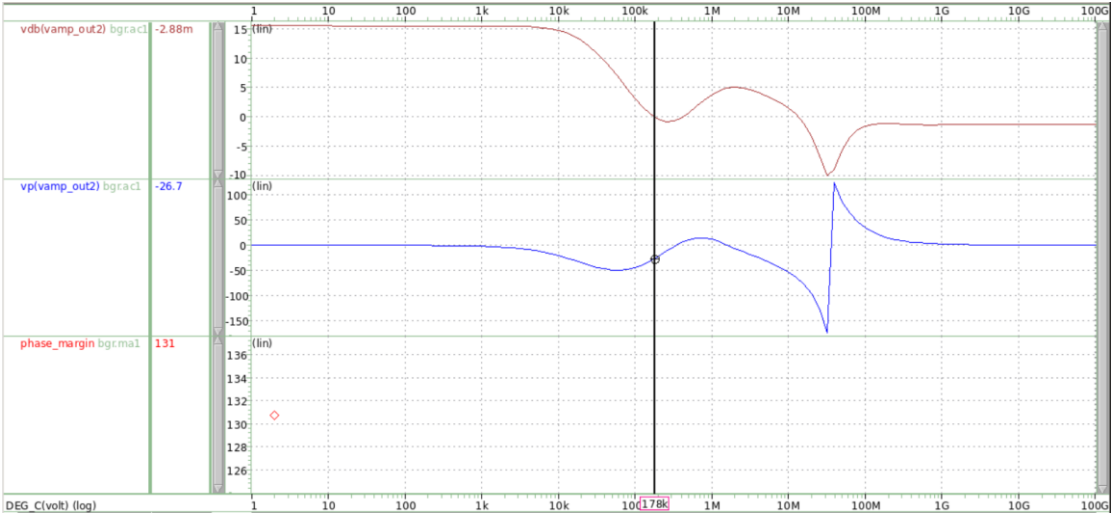
SS



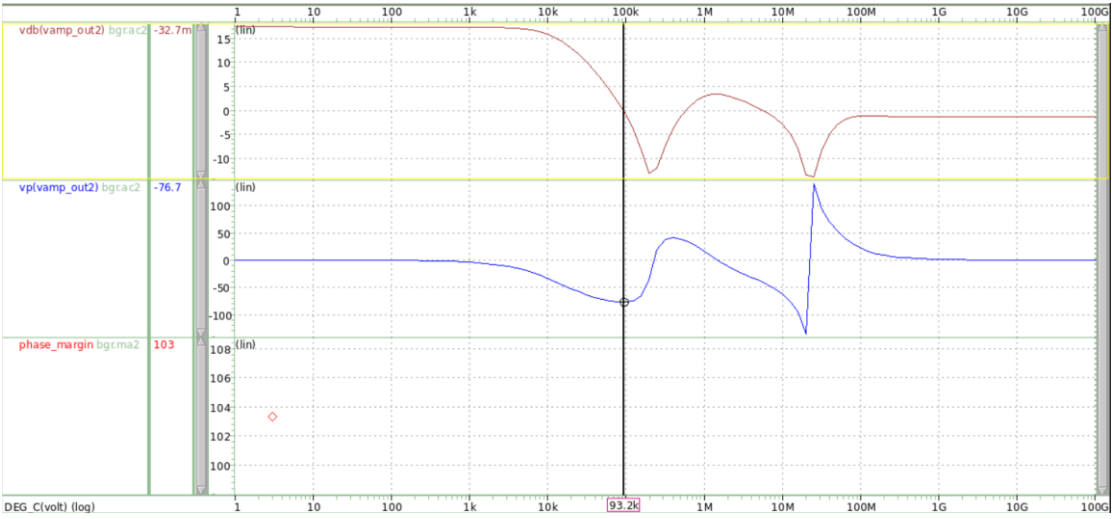
(c) .AC
TT



FF

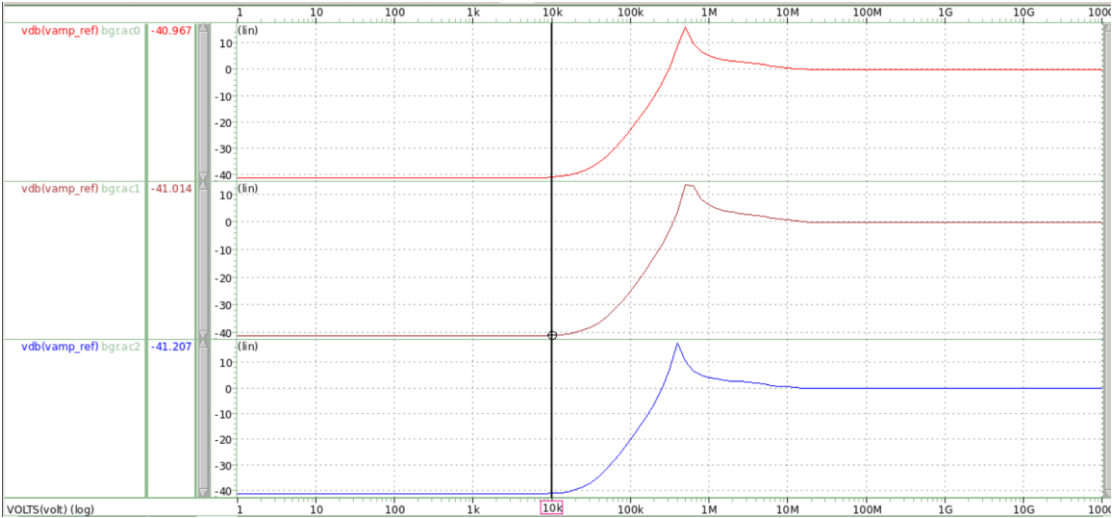
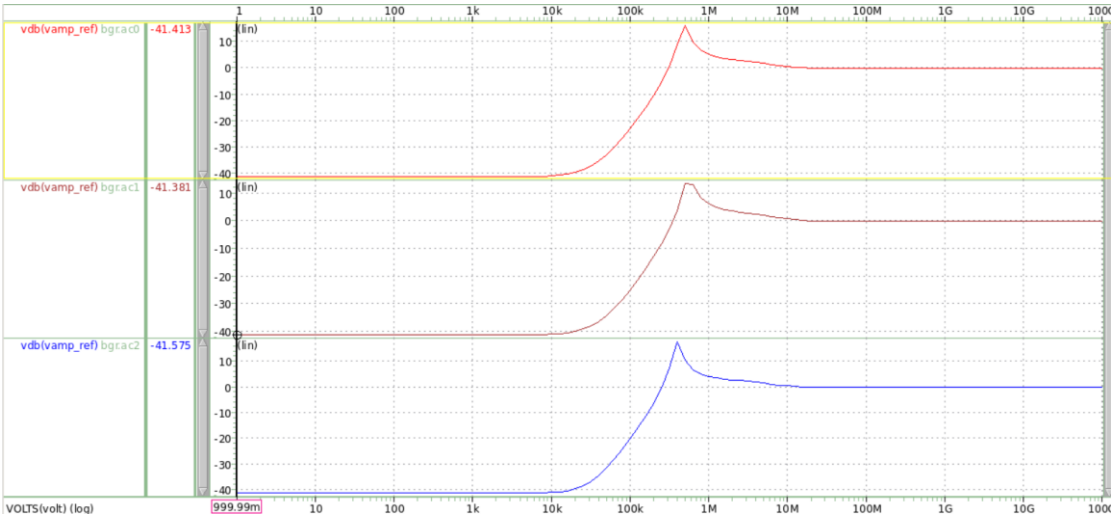


SS

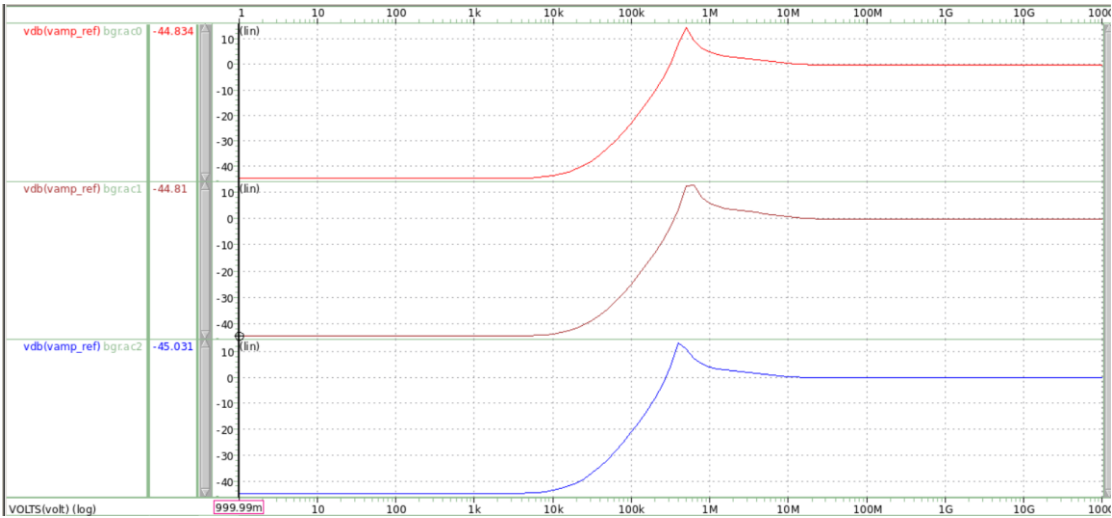


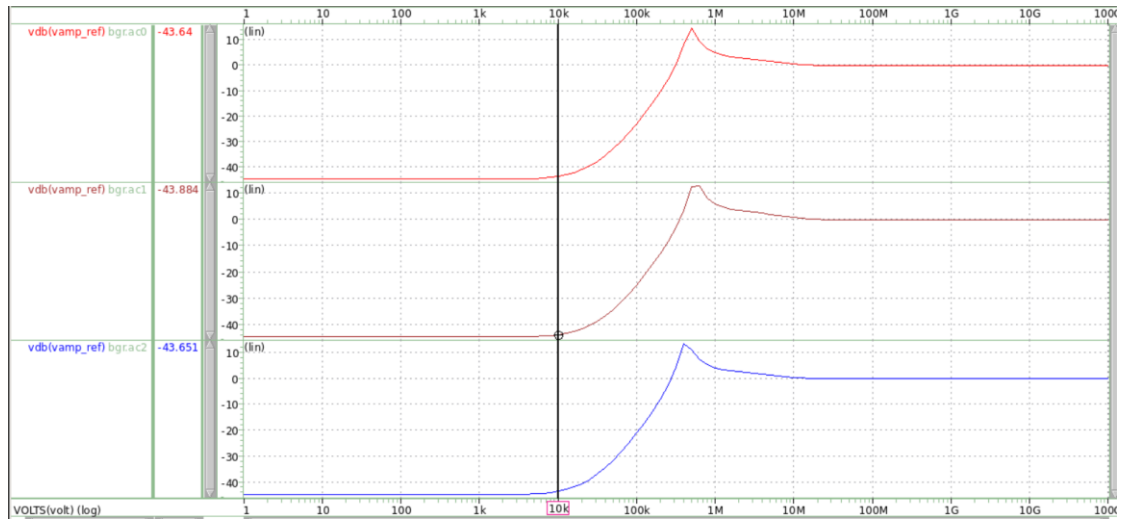
(圖由上到下為 TT FF SS)

125°C

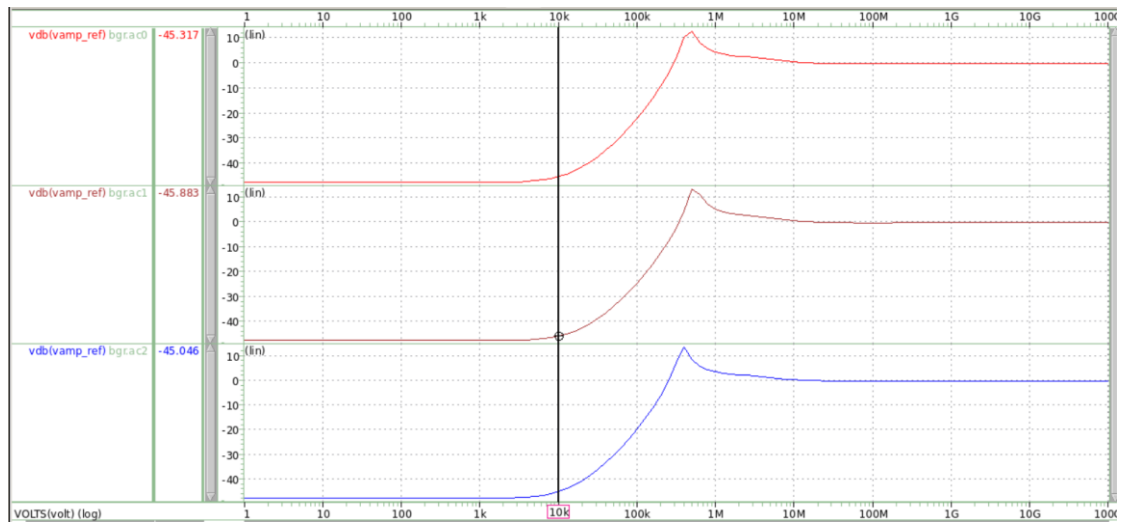
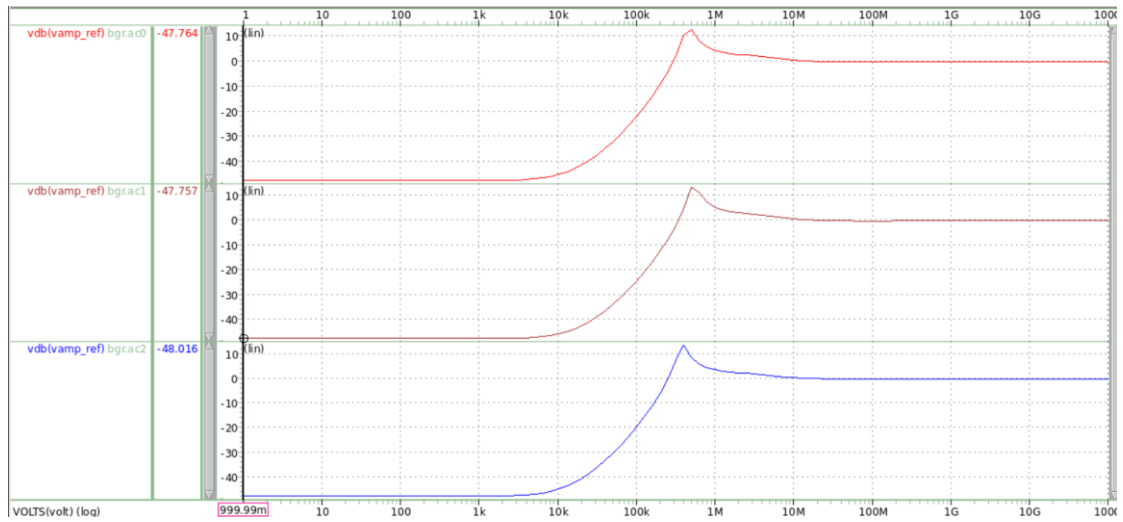


27°C





-40°C



3. SPICE code submission

Bandgap Reference							
Working Item	SPEC			Your Work			
Supply Voltage VDD	1.98V	1.8V	1.62V		1.98V	1.8V	1.62V
T.C. from -40~125°C	< 20ppm/°C			TT	8.77	7.51	11.37
				SS	8.65	7.41	11.39
				FF	8.81	7.60	11.35
Bandgap Voltage (V)	V			TT	1.36		
VDD	1.8V, -40~125°C						
PSR @ DC	< -40dB			TT	-41.4~-47.8		
				SS	-41.4~-47.8		
				FF	-41.5~-48.0		
PSR @ 10KHz	< -30dB			TT	-41.0~-45.3		
				SS	-41.0~-45.8		
				FF	-41.2~-45.0		
VDD	1.8V, 27°C, TT						
Power Consumption (μW)	< 50μW			36.41			

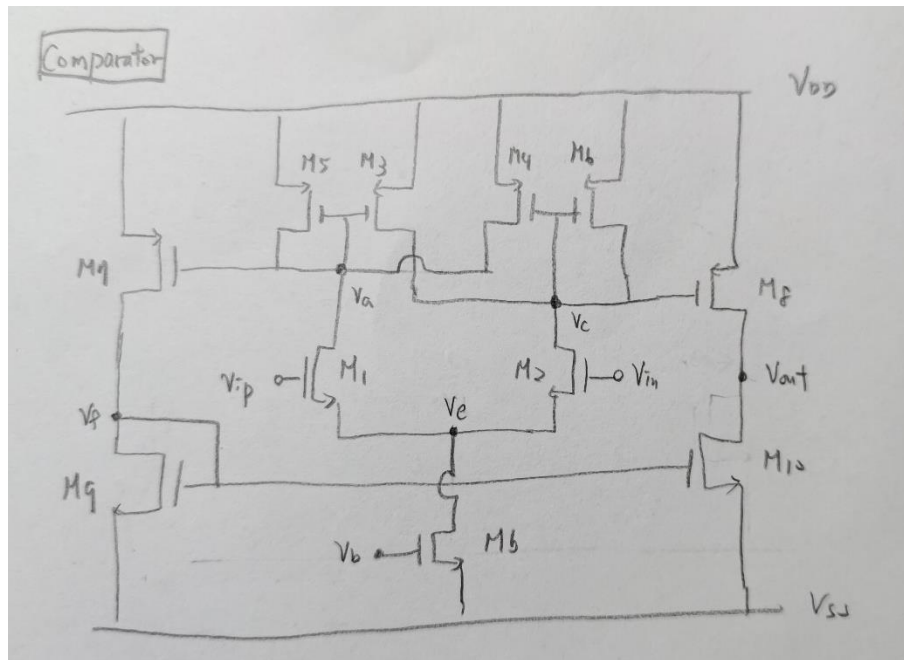
total voltage source power dissipation= 36.4071u watts

PART II – Single Slope ADC (SSADC)

(a) Please observe the operation of the SSADC and discuss the operational requirements that the comparator must meet (e.g. settling time, input range...).

由 SSADC 的流程圖中，可以發現到我的 comparator 的 settling time 要越小越好，這樣在高頻中讀取值的速度才快又精準，我們要求 20kHz 來取樣，因此 settling time 要遠小於 50u 秒，讀出來的狀態才會比較正確，而在 input range 中我要設 0.34-1.28V 的區間，大概差 0.94V，差距蠻大的，因此我在下面要設計時要盡量設計出高低電壓分明的 comparator 來降低誤差，而 comparator 中的 M3M4M5M6 是用來降低雜訊的設計，以免在交界處有雜訊時，而產生出大量的震動。

(b) Complete the design of the comparator (An opamp connected with an inverter chain) and describe your design in detail, including the schematic, operation points of all the transistors (you should draw your schematic and mark the DC voltage and current on it), and how you run the simulations. For you reference, a basic Please observe the operation of the SSADC and discuss the operational requirements that the comparator must meet (e.g. settling time, input range...). implementation is shown in Fig. 6.



在 comparator 的設計中，我是先設計 L 固定在 5u，電流要大這樣 slew rate 變化才會比較快，因此設計 Mb 的 W 設計成比較大的 6u，而因為 PMOS 比 NMOS 的流電能力還小，因此我 M7M8 設的 W 為 M9M10 的兩倍，這樣 comparator 的 Vin 在 1V 左右的地方變化才会有明顯的直角的效果，也就是當 Vin 電壓一變化經過判斷電壓時 Vout 的電壓變化會非常劇烈。

subckt						
element	0:m1	0:m2	0:m3	0:m5	0:m4	0:m6
model	0:n_18.1	0:n_18.1	0:p_18.1	0:p_18.1	0:p_18.1	0:p_18.1
region	Subth	Subth	Subth	Subth	Subth	Subth
id	8.2252p	8.2252p	-4.6355p	-4.6355p	-4.6347p	-4.6347p
ibs	-159.7681a	-159.7681a	1.212e-27	1.212e-27	1.212e-27	1.212e-27
ibd	-242.0975a	-242.0986a	17.7532a	17.7538a	17.7538a	17.7532a
vgs	198.5909m	198.5909m	-145.6200m	-145.6200m	-145.6145m	-145.6145m
vds	412.9709m	412.9764m	-145.6145m	-145.6200m	-145.6200m	-145.6145m
vbs	-801.4091m	-801.4091m	0.	0.	0.	0.
vth	479.5216m	479.5216m	-468.1096m	-468.1096m	-468.1096m	-468.1096m
vdsat	38.4995m	38.4995m	-39.8526m	-39.8526m	-39.8526m	-39.8526m
vod	-280.9307m	-280.9306m	322.4895m	322.4895m	322.4951m	322.4951m
beta	59.4063u	59.4063u	14.3095u	14.3095u	14.3095u	14.3095u
gam_eff	526.3234m	526.3234m	557.0847m	557.0847m	557.0847m	557.0847m
gm	316.2508p	316.2513p	134.1623p	134.1624p	134.1412p	134.1411p
gds	2.1968p	2.1968p	619.6217f	619.5501f	619.4514f	619.5229f
gmb	48.8635p	48.8636p	46.0979p	46.0979p	46.0907p	46.0907p
cdtot	1.3288f	1.3288f	1.4586f	1.4586f	1.4586f	1.4586f
cgtot	7.4157f	7.4157f	11.4149f	11.4149f	11.4150f	11.4150f
cstot	1.4111f	1.4111f	1.5378f	1.5378f	1.5378f	1.5378f
cbtot	8.6013f	8.6013f	12.9721f	12.9721f	12.9721f	12.9721f
cgs	388.9513a	388.9513a	361.1014a	361.1014a	361.1011a	361.1011a
cgd	388.2976a	388.2976a	358.7072a	358.7072a	358.7072a	358.7072a

element	0:m7	0:m8	0:m9	0:m10	0:mb
model	0:p_18.1	0:p_18.1	0:n_18.1	0:n_18.1	0:n_18.1
region	Subth	Subth	Subth	Subth	Subth
id	-8.6102p	-8.6088p	11.0889p	11.0877p	14.0702p
ibs	1.884e-27	1.884e-27	-4.623e-27	-4.623e-27	-3.556e-27
ibd	237.5565a	237.5647a	-12.0375a	-12.0285a	-552.8342a
vgs	-145.6200m	-145.6145m	60.3809m	60.3809m	0.
vds	-1.2996	-1.2997	60.3809m	60.3357m	801.4091m
vbs	0.	0.	0.	0.	0.
vth	-467.4967m	-467.4967m	339.7472m	339.7473m	324.4762m
vdsat	-38.8081m	-38.8081m	35.5354m	35.5354m	35.5597m
vod	321.8767m	321.8822m	-279.3663m	-279.3665m	-324.4762m
beta	28.8822u	28.8822u	58.1317u	58.1317u	352.5405u
gam_eff	557.0847m	557.0847m	507.4459m	507.4459m	507.4459m
gm	249.4856p	249.4463p	399.5541p	399.5113p	512.4333p
gds	104.4441f	104.4270f	26.2291p	26.2867p	2.1416p
gmb	86.4323p	86.4189p	92.7770p	92.7673p	120.8302p
cdtot	2.2066f	2.2066f	1.6663f	1.6663f	7.8996f
cgtot	22.8245f	22.8246f	10.0881f	10.0881f	61.9723f
cstot	2.8943f	2.8943f	1.7009f	1.7009f	9.5014f
cbtot	25.0468f	25.0468f	11.8997f	11.8998f	70.0501f
cgs	722.3027a	722.3020a	389.6055a	389.6055a	2.3326f
cgd	717.4000a	717.4000a	388.3153a	388.3153a	2.3298f

下面是在跑模擬所用的數據，其中我的 sample 的點是 256

```
.param VSS=0
.param Fs=20k
.param sampling duty cycle=0.2
.param cnt freq=1.594x
*.param NFIN=511
*.param NSAMPLE=1024

.param NFIN=119
.param NSAMPLE=256

.param cnt period='1/cnt freq'
.param Ts='1/Fs'

***** you can modify start*****
.param VDD=1.98 $ you could only have three type of
VDD(1.62V, 1.8V 1.98V)
$ There is no spec when
supply voltage=1.62V and 1.98V
$ just for you to observe the
performance change of ssAD

.param Vb=0.45 $ comparator biasing voltage

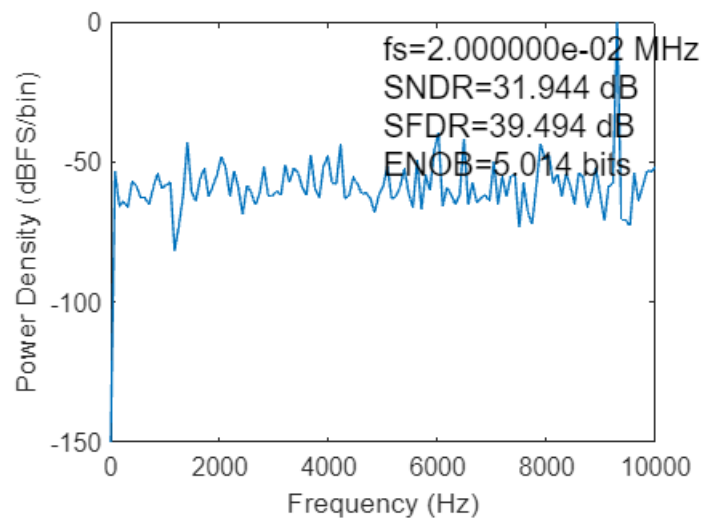
.param ramp max = 1.28 $ ramping signal maximum value
.param ramp min = 0.34 $ ramping signal minimum value

.param VCM = 0.81 $ common mode of input sine wave
.param amplitude = 468m $ amplitude of input sine wave " (ramp max-
ramp min)/2 - 2mV "
***** you can modify end*****
```

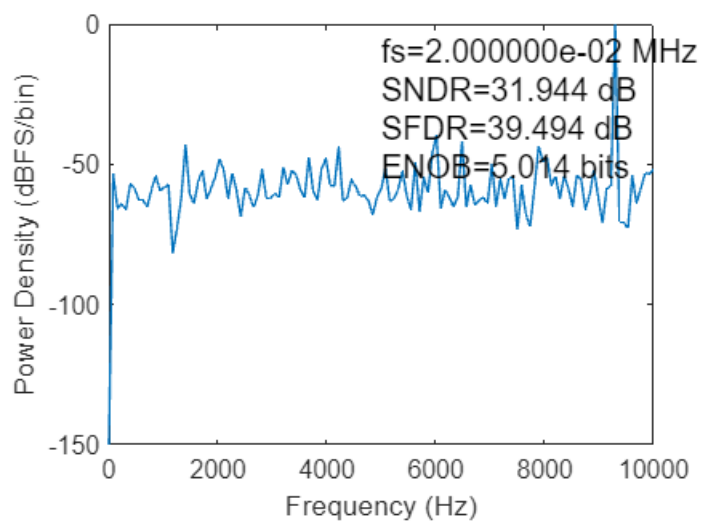
2. SPICE code submission

3. Measure the ENOB

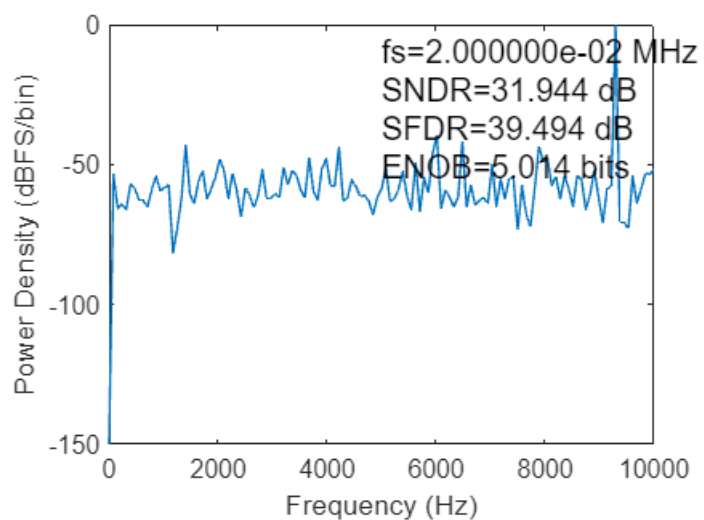
1.8V



1.68V



1.92V



Bandgap Reference + SSADC						
Working Item	SPEC			Your Work		
Single Slope ADC input range	0.95Vref~0.25Vref (Round to the second decimal place.)					
Supply Voltage VDD	1.8V	1.62V	1.98V	1.8V	1.62V	1.98V
Sampling rate	20kHz					
ENOB	> 5bits			5.014	5.014	5.014

基本上在不同電壓下，我所測出來的結果一模一樣，這代表前面設計的 Vref 很穩定，不太會因為電壓改變有太多的變化或雜訊。

PART III – Experience Sharing

這學期在這一堂課讓我深深地複習之前學的電子學，也讓我知道之前電子學的知識怎麼真實的應用在真實的設計電路之中，我在設計和調整數據時，我覺得還蠻有趣的，可以知道每個參數背後會影響到甚麼，只是最後寫報告有點麻煩，但可以趁機練習打字，但希望以後可以少打一點，最後謝謝助教辛苦的出題目，我感覺出題目很麻煩，阿我 HW2 因為交錯檔案而零分，可以因為我交了 sp 檔加一點分嗎。